Reference Material

Getting Printer Driver Info and Generating Output Image

August 31, 2017
Dai Nippon Printing Co., Ltd.

The copyrights for this document are the property of the holders of rights. Reproduction of any or all of the contents is prohibited.

The contents of this document are subject to change without prior notice.

Microsoft and Windows are registered trademarks of Microsoft Corporation valid in the USA and other countries.

1. Get Paper Size

You can get the paper size by getting the print object's ScaleWidth and ScaleHeight. By changing ScaleMode you can switch unit type to twip, mm, inch, or pixel. You can also get the paper size using GetDeviceCaps().

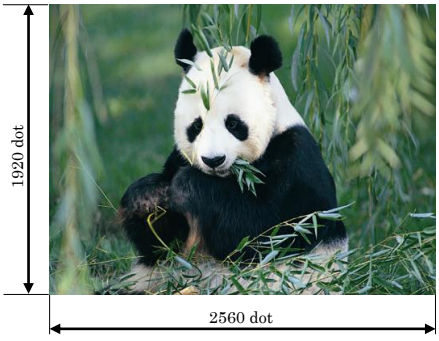
2. Get Resolution

You can get the resolution from dmPrintQuality and dmYresolution in the DEVMODE structure.

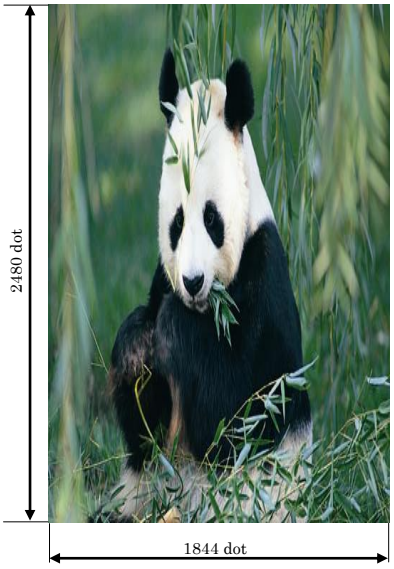
dmPrintQuality : x-axis resolution (head width resolution)
dmYresolution : y-axis resolution (paper feed direction resolution)

3. Generating Output Image

A printer has a horizontal resolution of 300dpi, and if a vertical resolution is set as 600dpi, when generating an output image, it is necessary to generate an image that is stretched in the paper feed direction, as shown below. (Refer to 3-1 Generation Procedure)



[Original image (5.0 mega pixels)]



[Generated image (6x4 size)]

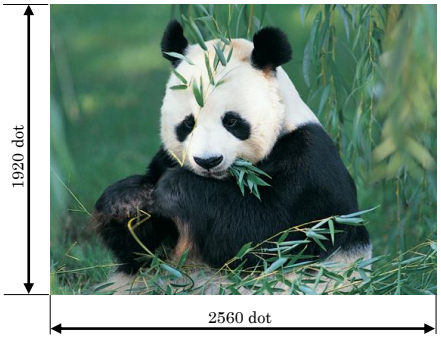
3-1 Generation Procedure (6x4 size, In the case of vertical resolution of 600dpi)

(1) First we get the pixel size for a horizontal and vertical resolution of 600dpi.
In the case of PC size, the output image is 1844(W) x 2480(H) dot. However, the size if the horizontal resolution were 600dpi would be

x = 1844 x (600 / 300)
= 3688 dot

So with both resolutions at 600dpi, the pixel size should become 3688(W) x 2480(H) dot.

(2) In order to make the pixel size for 600dpi resolution 3688(W) x 2480(H) dot, we have to resize the original image (with a fixed W / H ratio). Doing this, the size is resized to larger than 3688 x 2480dot, and the excess is trimmed off.
In the case of the illustration below, since the original image is 2560 x 1920dot, it has to be enlarged 144.1% (3688 / 2560) in order to become 3688 x 2480dot. Doing this, the Height becomes 2760dot (1920 x1.441), so to make it 2480dot, the excess at the top and bottom has to be trimmed off. (Depending on the image, the left and right sides may be trimmed instead.)



[Original image (5.0 mega pixels)]

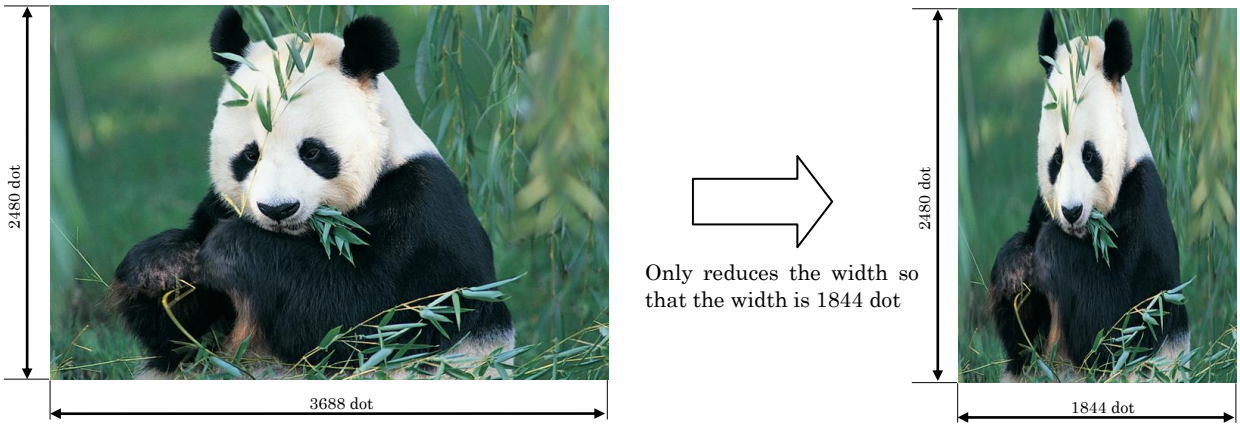
Enlarged 144.1%



Top and bottom are trimmed



(3) Then we make only the horizontal resolution 300dpi.
We resize (reduce) the 3688(W) x 2480(H) dot image so that only the width becomes 1844 dot.
With the following procedure, we have completed generation of the output image.



[Generated image]

■ Remarks
In the method above, first we enlarge the image to make the size 600dpi x 600dpi, then reduce the width. Another method is to reduce the image to make the size 300dpi x 300dpi, then enlarge the height to generate the output image. However, enlarging a reduced image may cause the image to deteriorate. Therefore, we think that enlarging the image to 600dpi x 600dpi, then reducing it is preferable.

Reference – image size

Paper	Paper Orientation	Image size (width x height)		Image pixel size (width x height)		
		(inch)	(mm)	300x300 dpi	300x600 dpi	600x600 dpi
5x3.5 (L)	Portrait	5"x 3.5"	131.1 x 92.1	1548 x 1088	1548 x 2176	3096 x 2176
	Landscape	3.5"x 5"	92.1 x 131.1	1088 x 1548	2176 x 1548	2176 x 3096
6x4 (PC)	Portrait	6"x 4"	156.1 x 105.0	1844 x 1240	1844 x 2480	3688 x 2480
	Landscape	4"x 6"	105.1 x 156.1	1240 x 1844	2480 x 1844	2480 x 3688
5x5	Portrait	5"x 5"	131.1 x 130.4	1548 x 1540	1548 x 3080	3096 x 3080
	Landscape	5"x 5"	130.4 x 131.1	1540 x 1548	3080 x 1548	3080 x 3096
6x6	Portrait	6"x 6"	156.1 x 155.4	1844 x 1836	1844 x 3672	3688 x 3672
	Landscape	6"x 6"	155.4 x 156.1	1836 x 1844	3672 x 1844	3672 x 3688
5x7 (2L)	Portrait	5"x 7"	131.1 x 181.0	1548 x 2138	1548 x 4276	3096 x 4276
	Landscape	7"x 5"	181.0 x 131.1	2138 x 1548	4276 x 1548	4276 x 3096
6x8 (A5)	Portrait	6"x 8"	156.1 x 206.2	1844 x 2436	1844 x 4872	3688 x 4872
	Landscape	8"x 6"	206.2 x 156.1	2436 x 1844	4872 x 1844	4872 x 3688
PR_3.5x5	Portrait	3.5"x 5"	92.1 x 131.1	1088 x 1548	2176 x 1548	2176 x 3096
	Landscape	5"x 3.5"	131.1 x 92.1	1548 x 1088	1548 x 2176	3096 x 2176
PR_4x6	Portrait	4"x 6"	105.1 x 156.1	1240 x 1844	2480 x 1844	2480 x 3688
	Landscape	6"x 4"	156.1 x 105.0	1844 x 1240	1844 x 2480	3688 x 2480
White border T/B		Top & Bottom 4.5 mm each		Top and Bottom 54 pixel each (6x4:60 Pixel)	Top and Bottom 108 pixel each (6x4:120 Pixel)	Top and Bottom 108 pixel each (6x4:120 Pixel)
White border L/R		Left & Right 5.0 mm each		Left and Right 60 pixel each	Left and Right 60 pixel each	Left and Right 120 pixel each

When set to "Border", the border is subtracted from the image pixel sizes shown above.

(*)The print area size is set larger than the actual paper size (1.5mm larger top, bottom, 2.0mm larger left, right).